

Fast Food Inventory Order Planner: Workbook Logic

1. Project Overview

The Fast-Food Inventory Order Planner is an Excel-based operational planning workbook designed to calculate recommended order quantities for inventory items in a fast-food or retail store environment. The workbook uses current stock, incoming ordered stock, delivery schedule, expected demand, usage rates, and buffer requirements to determine whether each item needs to be ordered.

This is a portfolio version of the workbook. It uses anonymised item names and synthetic sales data, so it does not contain confidential business, supplier, customer, or workplace information.

The main purpose of the workbook is to reduce the risk of stock shortages while also avoiding unnecessary over-ordering. It provides a structured and automated approach to stock planning, making the ordering process more consistent, transparent, and easier to manage.

2. Workbook Structure

The workbook is divided into six main sheets:

1. Instructions

This sheet explains how to use the workbook, which fields should be edited, and how to interpret the output.

2. Order_Planner

This is the main user-facing sheet. It allows the user to enter current stock and incoming ordered quantities, then automatically calculates the recommended order quantity and item status.

3. Data_Sales

This sheet contains synthetic historical sales data. The data is used to support demand forecasting and demand profile selection.

4. Usage_Rates

This sheet stores expected item usage by weekday and demand profile. Each item has usage assumptions for Low, Normal, and High demand conditions.

5. Setup

This sheet contains the planning rules used by the workbook, including delivery days, delivery coverage flags, weekday lists, demand profile multipliers, and profile rules.

6. Forecast_Model

This sheet calculates forecasted weekly demand, compares it with historical baseline sales, and helps determine the selected demand profile.

3. Instructions Sheet Logic

The Instructions sheet provides guidance for the end user. It explains the purpose of the workbook and the steps required to use the planner correctly.

The user is instructed to review the planning inputs, enter current stock values, enter any incoming ordered stock, and then check the recommended order quantity. The sheet also explains that formula cells should not be edited.

The instructions clarify that missing dates in the sales table represent store-closed days and are intentionally excluded from demand forecasting. This prevents closed days from incorrectly reducing the sales baseline.

The sheet also states that the workbook is a portfolio version using anonymised products and synthetic sales data.

4. Order_Planner Sheet Logic

The Order_Planner sheet is the main operational sheet in the workbook. It brings together the planning inputs, item-level stock data, demand assumptions, and final order recommendations.

The key inputs on this sheet are:

- Stocktake Day
- Delivery Day
- Include Delivery Day in Pre-Demand
- Demand Profile
- Current Stock
- Incoming Ordered

The key calculated outputs are:

- Pre-Delivery Demand
- Coverage Demand
- Adjusted Buffer
- Total Required
- Order Qty

- Status
- Stock Runway
- Projected Stock at Delivery

The user mainly enters stock-related information, while the workbook automatically calculates the required order quantity for each item.

5. Planning Input Logic

The Stocktake Day represents the day on which physical stock is counted. The Delivery Day represents the next day on which delivery is expected to arrive.

The Include Delivery Day in Pre-Demand option controls whether demand on the delivery day should be counted before delivery arrives. If the delivery arrives after opening or the delivery time is uncertain, this option should be set to TRUE. This is the safer setting because it assumes the store may need to operate for part of the delivery day before the new stock is available.

The Demand Profile represents the expected demand level for the planning period. The workbook uses forecasted sales and profile rules to support demand profile selection.

6. Usage_Rates Sheet Logic

The Usage_Rates sheet stores the expected usage of each inventory item by weekday and demand profile.

Each item has three demand profiles:

- Low
- Normal
- High

For each profile, the workbook stores expected usage for Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, and Sunday. This allows the planner to calculate demand more accurately based on the selected day and expected demand level.

The basic weekly usage logic is:

Weekly Usage = Monday Usage + Tuesday Usage + Wednesday Usage + Thursday Usage + Friday Usage + Saturday Usage + Sunday Usage

The workbook also includes a buffer value for each item. This buffer is used to reduce the risk of running out of stock due to unexpected demand changes or operational uncertainty.

7. Demand Profile Multiplier Logic

The workbook uses demand profile multipliers to adjust demand levels. A higher multiplier increases the expected stock requirement, while a lower multiplier keeps the requirement closer to the base usage level.

For example:

- Low demand may use a lower or base multiplier.
- Normal demand may use a moderate multiplier.
- High demand may use a higher multiplier.

The adjusted buffer is calculated using the selected profile multiplier.

Adjusted Buffer = Buffer × Profile Multiplier

The workbook rounds the adjusted buffer upward so that stock requirements are not understated.

For example, if the buffer is 4 and the profile multiplier is 1.15, the adjusted buffer becomes 4.6. If rounded up, the adjusted buffer becomes 5.

8. Setup Sheet Logic

The Setup sheet controls the main planning rules used in the workbook.

It includes:

- Delivery Days
- Delivery Coverage Flags
- Weekday List
- Demand Profile Multipliers
- Profile Rules

The Delivery Days table defines the valid delivery days. These values are used in the Delivery Day dropdown on the Order_Planner sheet.

The Delivery Coverage Flags table defines which weekdays each delivery needs to cover. For example, a Monday delivery may need to cover Tuesday and Wednesday demand until the next delivery cycle.

The Demand Profile table stores the profile multipliers used to adjust demand.

The Profile Rules table defines the conditions used to classify the weekly forecast into Low, Normal, or High demand.

9. Delivery Coverage Logic

Delivery coverage determines which days of demand the order should cover after delivery arrives.

For example, if the next delivery is on Monday and the next ordering cycle needs to cover Tuesday and Wednesday, the coverage demand is calculated as:

Coverage Demand = Tuesday Usage + Wednesday Usage

The workbook uses delivery coverage flags from the Setup sheet to decide which weekday usage values should be included in the coverage demand calculation.

This approach makes the workbook more flexible because delivery rules can be adjusted without rewriting the main order formulas.

10. Data_Sales Sheet Logic

The Data_Sales sheet contains synthetic historical sales data. The data is used to calculate historical baseline sales and support the forecast model.

The main columns include:

- Date
- Weekday
- Sales
- YearWeek
- Week Type

The Date column records the sales date. The Weekday column identifies the weekday for each date. The Sales column contains synthetic daily sales values. The YearWeek column supports weekly grouping and analysis.

Missing dates in this sheet represent store-closed days. These dates are intentionally excluded from demand forecasting instead of being treated as zero-sales trading days. This prevents the forecast from being reduced incorrectly.

11. Forecast_Model Sheet Logic

The Forecast_Model sheet compares forecasted weekly sales with historical baseline sales.

The workbook calculates forecasted sales for each weekday and compares them with historical weekday averages. This helps estimate whether expected demand is lower, normal, or higher than usual.

The weekly forecast is calculated as:

Weekly Forecast = Monday Forecast + Tuesday Forecast + Wednesday Forecast + Thursday Forecast + Friday Forecast + Saturday Forecast + Sunday Forecast

The historical baseline is calculated using past sales data. The workbook compares similar weekdays with each other, such as Monday with previous Mondays and Saturday with previous Saturdays.

The weekly multiplier is calculated as:

Weekly Multiplier = Weekly Forecast / Weekly Baseline

For example, if the weekly forecast is 42,000 and the historical baseline is 40,000, the weekly multiplier is:

Weekly Multiplier = 42,000 / 40,000 = 1.05

This means the forecasted demand is 5% higher than the historical baseline.

12. Demand Profile Selection Logic

The weekly multiplier is used to select the demand profile.

For example:

- If the weekly multiplier is low, the selected profile may be Low.
- If the weekly multiplier is close to the normal range, the selected profile may be Normal.
- If the weekly multiplier is high, the selected profile may be High.

The selected profile then feeds into the Order_Planner sheet and Usage_Rates sheet. This ensures that the order recommendation reflects the expected demand level.

13. Pre-Delivery Demand Logic

Pre-Delivery Demand estimates how much stock will be used before the next delivery arrives.

This is important because the store must have enough stock to operate between the stocktake time and the delivery arrival time.

For example, if the stocktake is completed on Sunday and delivery is expected on Monday, and they Include Delivery Day in Pre-Demand option is TRUE, the workbook includes both Sunday and Monday demand.

Pre-Delivery Demand = Demand from Stocktake Day to Delivery Day

If delivery-day demand is included, the formula assumes that the delivery may arrive after opening or at an uncertain time. This helps prevent under-ordering.

14. Projected Stock at Delivery Logic

Projected Stock at Delivery estimates how much stock will remain by the time the delivery arrives.

The formula is:

Projected Stock at Delivery = Current Stock + Incoming Ordered - Pre-Delivery Demand

For example:

Current Stock = 10

Incoming Ordered = 0

Pre-Delivery Demand = 6

Projected Stock at Delivery = $10 + 0 - 6 = 4$

This means the store is expected to have 4 units remaining when the delivery arrives.

If the projected stock is zero or negative, the item may run out before or around the delivery time.

15. Coverage Demand Logic

Coverage Demand estimates how much stock is required after the delivery arrives.

This is based on the delivery coverage flags in the Setup sheet. The workbook identifies which weekdays the delivery needs to cover and then adds the expected item usage for those days.

For example:

Delivery Day = Monday

Coverage Days = Tuesday and Wednesday

Coverage Demand = Tuesday Usage + Wednesday Usage

This ensures that the recommended order quantity is based not only on current stock, but also on the demand expected after delivery.

16. Total Required Logic

Total Required represents the amount of stock needed to cover expected demand and buffer requirements.

The formula is:

Total Required = Coverage Demand + Adjusted Buffer

For example:

Coverage Demand = 12

Adjusted Buffer = 3

Total Required = $12 + 3 = 15$

This means the item should have 15 units available to safely cover demand and buffer requirements.

17. Order Quantity Logic

The Order Qty column calculates the recommended quantity to order for each item.

The formula is:

Order Qty = $\text{MAX}(0, \text{Total Required} - \text{Projected Stock at Delivery})$

This means the workbook only recommends an order if the required quantity is greater than the stock expected to be available.

For example:

Total Required = 15

Projected Stock at Delivery = 4

Order Qty = $15 - 4 = 11$

If projected stock is already greater than total required, the order quantity becomes 0.

For example:

Total Required = 15

Projected Stock at Delivery = 18

Order Qty = $\text{MAX}(0, 15 - 18) = 0$

This prevents the workbook from recommending negative order quantities.

18. Status Logic

The workbook classifies each item into one of three statuses: URGENT, ORDER, or OK.

URGENT means the projected stock at delivery is zero or negative. This indicates that the item may run out before or around the next delivery.

ORDER means the item is not immediately critical, but additional stock is still recommended.

OK means current and incoming stock are sufficient to cover expected demand and buffer requirements.

The status logic helps users quickly prioritise items. URGENT items should be reviewed first because they represent the highest stockout risk.

19. Stock Runway Logic

Stock Runway estimates how long the current stock may last based on expected usage.

A simplified version of the logic is:

Stock Runway = Current Available Stock / Average Daily Usage

For example:

Current Stock = 12

Average Daily Usage = 4

Stock Runway = $12 / 4 = 3$ days

This helps the user understand how quickly each item may run out under expected demand conditions.

20. Formula Protection Logic

The workbook uses sheet protection to reduce the risk of accidental formula changes.

User-editable cells are limited to key input fields such as:

- Stocktake Day
- Delivery Day
- Include Delivery Day in Pre-Demand
- Current Stock
- Incoming Ordered

Formula cells are protected because they calculate the order recommendations and stock status.

This design makes the workbook safer for practical use because users can update operational inputs without damaging the calculation logic.

21. Data Validation Logic

The workbook uses dropdown lists to reduce input errors.

For example:

- Stocktake Day uses a weekday dropdown.
- Delivery Day uses a valid delivery day dropdown.
- Include Delivery Day in Pre-Demand uses a TRUE/FALSE dropdown.
- Demand Profile is linked to the forecast and profile rules.

This prevents spelling errors and inconsistent entries. It also makes the workbook easier for non-technical users to operate.

22. Why the Workbook Separates Pre-Delivery and Coverage Demand

The workbook separates demand into two stages.

The first stage is before delivery arrives. This checks whether the store has enough stock to continue operating until delivery.

The second stage is after delivery arrives. This checks how much stock is needed to cover the next delivery cycle.

This separation makes the workbook more realistic than a simple reorder-point calculation. It considers both immediate stockout risk and future coverage requirements.

23. Confidentiality and Portfolio Version

The workbook has been prepared as a portfolio version.

It uses:

- Anonymised item names
- Synthetic sales data
- Sample stock values
- Sample usage rates
- Sample forecast values
- No supplier names
- No customer data
- No company-specific financial data
- No confidential workplace information

This allows the project to demonstrate Excel, analytics, and operational planning skills without exposing private business information.

24. Skills Demonstrated

This workbook demonstrates the following skills:

- Microsoft Excel workbook design
- Excel Tables
- Structured references

- Data validation
- Conditional formatting
- Sheet protection
- Forecasting logic
- Inventory planning
- Operational analytics
- Formula-based automation
- Demand profiling
- Stock coverage planning
- User documentation
- Business process improvement

25. Summary

The Fast Food Inventory Order Planner automates the inventory ordering process by combining stock inputs, delivery timing, demand forecasting, usage rates, and buffer logic. It calculates whether each item is urgent, should be ordered, or is sufficiently stocked.

The workbook supports better operational decision-making by reducing manual calculation, improving consistency, and helping users identify stockout risks before they occur. As a portfolio project, it demonstrates how Excel can be used as a practical business analytics tool in a fast-paced retail or food service environment.